

NOVA CART BUSINESS INTELLIGENCE PROJECT

**End-to-End Business Intelligence
Solution using Excel, MySQL &
Power BI**

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1. Executive Summary

The Nova Cart Business Intelligence Project is an end-to-end analytics solution developed to turn raw e-commerce data into meaningful business insights that support strategic decision-making. The project follows a complete BI workflow like starting with raw data exploration in Microsoft Excel, moving into relational database management and analytical querying in MySQL, and finally wrapping up with interactive dashboard development in Microsoft Power BI.

The main goal of this project was to look closely at Sales Performance, Customer Intelligence, Product Performance, and Operational Efficiency. Through structured data modelling, SQL-based analysis, DAX calculations, and interactive dashboards, the project brings out insights that genuinely help in making better business decisions.

This portfolio project reflects the complete Business Intelligence lifecycle, bringing together data preparation, relational modelling, SQL analysis, KPI development, and executive reporting.

Project Information

Attribute	Details
Project Name	Nova Cart Business Intelligence Dashboard
Domain	E-Commerce Retail Analytics
Industry	Online Retail & E-Commerce
Project Type	End-to-End Business Intelligence Project
Tools Used	Microsoft Excel, MySQL, Microsoft Power BI
SQL Concepts	Joins, Aggregations, CASE, Window Functions, CTEs, Ranking, Subqueries
Power BI Features	Data Modeling, DAX Measures, KPIs, Interactive Dashboards
Dashboards Developed	4 Executive Dashboards

Business Goal

Develop a centralized Business Intelligence solution capable of transforming transactional data into executive-level insights. The solution enables stakeholders to monitor sales, customer behaviour, product performance and operational efficiency through interactive dashboards and data-driven reporting.

2. Business Problem, Project Objectives & Dataset Overview

Business Problem

Nova Cart operates in a competitive e-commerce environment where large volumes of transactional data make decision-making challenging. The objective of this project was to transform raw business data into meaningful insights using a complete Business Intelligence workflow.

Project Objectives

- Develop an end-to-end Business Intelligence solution using Excel, MySQL and Power BI.
- Build a relational data model using multiple interconnected tables.
- Perform SQL-based business analysis to answer sales, customer, product and operational questions.
- Create interactive Power BI dashboards with DAX measures and KPIs.
- Generate actionable insights for business decision-making.

Table	Purpose
Customers	Customer information
Orders	Order transactions
Order Items	Product level order details
Products	Product information
Categories	Category details
Brands	Brand details
Reviews	Customer ratings
Returns	Returned orders
Cities	Geographical analysis

Dataset Overview

The project is based on a relational e-commerce dataset consisting of nine interconnected tables covering customers, orders, order items, products, categories, brands, reviews, returns and cities.

Data Model

Primary and foreign key relationships were established across the tables to enable SQL joins, data integrity and Power BI data modeling. This unified model formed the foundation for all dashboards and business analyses.

3. Project Workflow & Methodology

End-to-End Business Intelligence Workflow

The Nova Cart project followed a structured Business Intelligence lifecycle, beginning with raw transactional data and ending with interactive executive dashboards. Each stage of the workflow contributed to transforming data into actionable business insights.

Workflow Stage	Description
Stage 1 – Data Collection	Raw e-commerce data containing customer, order, product, review, return, brand, category and city information was collected and reviewed.
Stage 2 – Microsoft Excel	Excel was used for initial data understanding, validation, exploration and preliminary analysis using tables, pivot tables and charts.
Stage 3 – MySQL	The dataset was imported into MySQL, relationships were established using primary and foreign keys, and business queries were written to analyze customer behavior, revenue, products and operations.
Stage 4 – Power BI	The cleaned data model was connected to Power BI where relationships were verified, DAX measures and KPIs were created, and four interactive dashboards were developed.
Stage 5 – Business Insights	The dashboards were used to identify trends, monitor KPIs and generate

	recommendations for business decision-making.
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Project Methodology

- Validated and explored raw data in Excel.
- Designed the relational database structure in MySQL.
- Performed SQL-based business analysis using joins, aggregations, CASE statements, ranking functions and window functions.
- Imported the analytical model into Power BI.
- Built DAX measures, KPI cards and interactive dashboards.
- Interpreted dashboard results to derive business insights and recommendations.

Deliverables

- SQL analytical queries
- Excel analysis workbook
- Four Power BI dashboards
- Business Intelligence report
- GitHub project repository

4. Sales Intelligence Dashboard Analysis

Dashboard Objective

The Sales Intelligence Dashboard provides an executive view of sales performance by consolidating revenue, order volume, customer activity and purchasing trends into a single analytical interface. It enables business users to monitor performance and support data-driven decision making.

Key Performance Indicators (KPIs)

KPI	Business Purpose
Total Revenue	Overall revenue generated
Total Orders	Total completed order transactions
Total Customers	Active customer base
Average Order Value	Average revenue per order

Key Visualizations

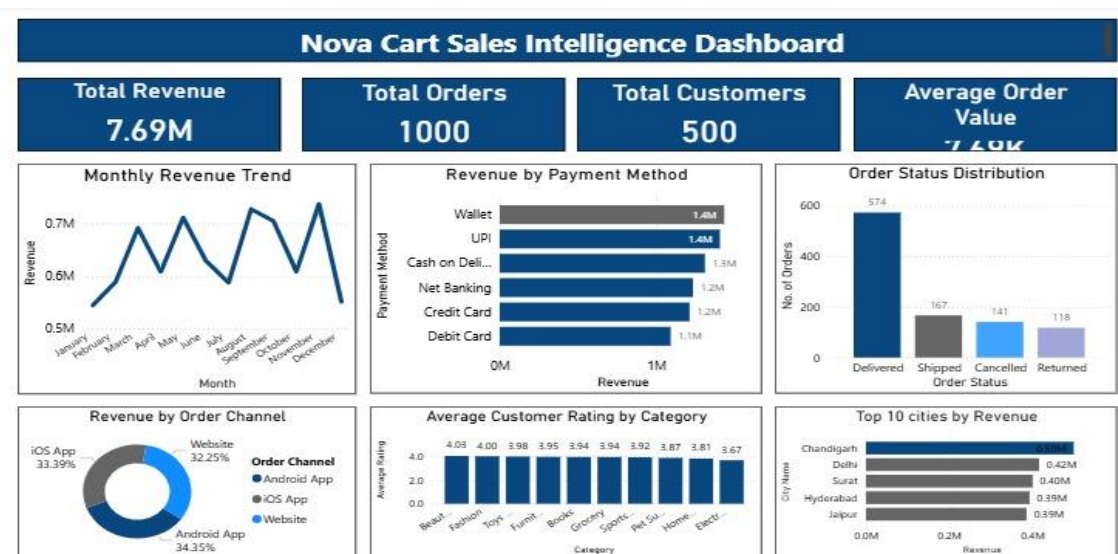
- Monthly Revenue Trend
- Revenue by Payment Method
- Top Cities by Revenue
- Revenue Distribution by Product Category

Key Business Insights

- Total revenue generated: **₹7.69M** across 1,000 orders from 500 customers, with an average order value of **₹7.68K**.
- **574 orders (57.4%)** were successfully delivered; 118 orders (11.8%) were returned.
- **Chandigarh, Delhi, and Surat** emerged as the top 3 revenue-generating cities.
- **Wallet and UPI** were the most preferred payment methods, contributing ₹1.4M each.
- Order channel split was nearly even: Android (34.35%), iOS (33.39%), Website (32.25%)

Management Recommendations

Prioritize high-performing markets, monitor monthly sales trends for demand planning, optimize promotional campaigns to improve average order value, and continue analyzing payment preferences to enhance the customer experience.



5. Customer Intelligence & Product Performance Dashboard

Customer Intelligence Dashboard

The Customer Intelligence Dashboard focuses on understanding customer behavior, loyalty, retention and acquisition patterns. It supports business decisions related to customer engagement, retention strategies and long-term customer value.

Key Customer KPIs

KPI	Business Purpose
Customer Retention Rate	Measures the percentage of customers who returned to place additional orders.
Repeat Customers	Tracks customers with multiple purchases.
Average Customer Rating	Monitors overall customer satisfaction.
Customer Lifetime Value	Estimates the average value generated by each customer.

Customer Insights

- Customer retention indicates the effectiveness of long-term engagement strategies.
- Customer retention rate stood at **66.5%**, with **290** repeat customers
- Average Customer Lifetime Value (CLV) was **₹17.6K** per customer
- Average customer rating across categories: **3.91/5**
- Android app led customer acquisition (**256 customers, 34.69%**), followed by iOS (250, 33.88%)



Product Performance Dashboard

The Product Performance Dashboard evaluates product quality, pricing and category-level performance. It enables management to identify high-performing products, monitor customer ratings and optimize product portfolio decisions.

Key Product KPIs

KPI	Business Purpose
Average Product Rating	Measures overall product quality based on customer reviews.
Top Product Revenue	Identifies the highest revenue-generating product.
Average Selling Price	Evaluates pricing across products.
Total Product Categories	Represents product portfolio diversity.

Product Insights & Recommendations

- Top-performing product generated ₹7,689K in revenue
- Average product rating: 3.9/5 across 10 product categories
- Average selling price stood at ₹7.69K



6. Operational Performance, Business Findings & Strategic Recommendations

Operational Performance Dashboard

The Operational Performance Dashboard evaluates fulfilment efficiency, delivery performance and order execution. It helps management monitor operational KPIs and identify opportunities to improve customer experience while reducing operational costs.

Operational KPIs

KPI	Business Purpose
Average Delivery Time	Measures the average number of days required to deliver an order.
Return Rate	Tracks the percentage of returned orders.
Delivered Orders	Measures successful order fulfilment.
Average Shipping Fee	Monitors shipping cost efficiency.
Average Items per Order	Evaluates basket size and purchasing behavior.

Operational Insights

- Average delivery time: 4.4 days
- Return rate: 10.0%
- Average shipping fee: ₹54.0
- Average items per order: 1.9



Overall Business Findings

- Sales, customer, product and operational data can be monitored through a unified Business Intelligence solution.
- Customer retention and repeat purchase metrics provide visibility into long-term customer value.
- Product ratings and category performance help identify improvement opportunities.
- Operational KPIs support continuous monitoring of delivery efficiency and returns.
- Interactive dashboards enable faster, data-driven decision making.

Strategic Recommendations

- Invest more in high-performing customer acquisition channels.
- Strengthen loyalty initiatives to increase repeat purchases.
- Monitor low-rated products and review pricing or quality.
- Reduce delivery delays and investigate the root causes of returns.
- Review dashboard KPIs regularly to support continuous business improvement

7. Conclusion, Skills Demonstrated & Future Scope

Project Conclusion

The Nova Cart Business Intelligence Project successfully demonstrates an end-to-end Business Intelligence workflow that transforms raw e-commerce data into meaningful business insights. By combining Microsoft Excel, MySQL and Power BI, the project establishes a structured analytical framework capable of supporting strategic and operational decision-making. The four dashboards collectively provide a comprehensive view of sales performance, customer behavior, product performance and operational efficiency.

Skills Demonstrated

- Microsoft Excel (Data Validation, Pivot Tables & Exploratory Analysis)
- MySQL (Data Modeling, Joins, Aggregations, CASE Statements, Window Functions & Business Queries)
- Power BI (Data Modeling, DAX Measures, KPIs & Interactive Dashboards)
- Business Intelligence & Data Visualization

- KPI Design and Performance Monitoring
- Analytical Thinking & Business Storytelling

Future Scope

- Integrate live data sources for real-time dashboard monitoring.
- Develop predictive models for sales forecasting and customer retention.
- Implement automated data refresh and scheduled reporting.
- Expand the solution with advanced segmentation and forecasting analytics.

Portfolio Summary

This portfolio project reflects practical experience in designing a complete Business Intelligence solution—from raw data preparation and SQL-based analysis to interactive Power BI dashboards and business reporting. It demonstrates the ability to translate business requirements into analytical solutions and communicate insights through executive-level dashboards.

Final Statement

The Nova Cart Business Intelligence Dashboard serves as a portfolio-ready analytics solution, demonstrating technical proficiency, business understanding and data-driven decision-making capabilities.